Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *MMLU* | ***Context:*** *Non-Arab Tourists and Expats in Eight Arab Countries*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both MMLU and the deployment use text output, satisfying basic modality match. However, MMLU's classification-accuracy metric over A/B/C/D tokens [Q49, Q56] cannot evaluate the open-ended, multi-perspective explanatory responses the deployment's tourist/expat assistant produces. Calibration evaluation [Q64–Q67] is a useful auxiliary signal — GPT-3's 24% confidence-accuracy gap [Q66] is informative — but does not address the form mismatch. With OF rated MODERATE, this is a real but partial concern: MMLU can score factual recall in MCQ form but not the explanatory output form the deployment expects.

ID	Evidence
Q49	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q56	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.” (p.6)
Q64	“We should not trust a model's prediction unless the model is calibrated, meaning that its confidence is a good estimate of the actual probability the prediction is correct.” (p.7)
Q65	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)
Q66	“Its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.” (p.7)
Q67	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019).” (p.7)

Strengths

- Text output modality matches deployment text interface
- Calibration metrics (RMS calibration error, confidence-accuracy correlation) [Q65, Q67] provide useful auxiliary signals on whether models 'know what they don't know' [Q16] — directly relevant to a tourist/expat assistant that should hedge on contested topics
- Zero-shot/few-shot evaluation regime [Q3] aligns with deployment's open-domain generalization requirement (Strength 4)

ID	Evidence
Q3	“We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings.” (p.1)
Q16	“Worryingly, we also find that GPT-3 does not have an accurate sense of what it does or does not know since its average confidence can be up to 24% off from its actual accuracy.” (p.2)
Q65	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)
Q67	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019).” (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Modality matches (text), but output granularity mismatches: MCQ token probability [Q56] vs. open-ended explanation expected by deployment users.

OF-2: Not applicable — text-based deployment, no TTS requirement documented.

OF-3: Tourist/expat target population is literate adult readers; literacy not a concern. INSUFFICIENT DOCUMENTATION on accessibility-specific output requirements.

OF-4: Form mismatch documented: MCQ scoring [Q49] cannot credit appropriately-hedged multi-perspective open-ended responses; the authors note their evaluation format 'is not identical to the format in which information is acquired during pretraining' [Q76].

ID	Evidence
Q49	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q56	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject.” (p.6)
Q76	“Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining.” (p.8)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Whether the deployed system generates free-form text or also offers MCQ-style confirmations (interface design not specified in YAML)

Requires Expert Verification

- Whether deployment evaluation will supplement MMLU MCQ accuracy with open-ended generation metrics (e.g., MENAValues-style prompting)

Remediation

Gap 1: MCQ accuracy metric cannot evaluate open-ended explanatory or perspective-acknowledging responses that the deployed assistant produces.

Recommendation: Add open-ended generation evaluation with rubric-based scoring (cultural appropriateness, multi-perspective acknowledgment, calibrated hedging on contested topics). Re-evaluate every 6–12 months given rapid Arabic-benchmark ecosystem growth [WEB-12].

ID	Evidence
WEB-12	Survey of Arabic benchmarks notes gaps in coverage of North Africa, Levant, Iraq even within the Arabic NLP ecosystem (https://arxiv.org/html/2510.13430v1)